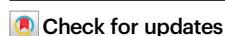


Non-equilibrium reducing flame aerosol process to create supported high-entropy alloy nanoparticles

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High-entropy alloy (HEA) nanomaterials provide opportunities and property combinations for energy and electronic applications, but their practical synthesis faces challenges of elemental immiscibility, metal reducibility, and particle aggregation during their synthesis. Herein, we report a broadly applicable non-equilibrium, scalable, and in-situ reducing flame aerosol process for synthesis of supported HEA nanoparticles. This versatile process can directly load a high concentration of 2–4 nm HEA nanoparticles on various 1- to 3-dimensional supports. Notably, simultaneous formation of HEA nanoparticles and a mesoporous silica support was successfully realized in a single step. Exploration of this process demonstrates the role of kinetics and entropy on decreasing alloy particle size and altering the reducibility of elements. We propose an entropy-induced reduction mechanism to incorporate oxidizable elements into HEAs, which extends the compositional space of HEA nanoparticles. As a representative catalytic application, we present a RuPdOsIrPt/graphene electrocatalyst with high activity and stability for hydrogen oxidation reaction. Our findings open horizons for high-performance HEA design and applications in diverse fields such as catalysis, electrochemistry, and sensing.

Alloys in which multiple elements randomly occupy sites in a single crystallographic lattice are known as solid solutions. In 2004, Yeh¹ and Cantor² et al. reported solid solution alloys containing 5 or 6 elements with near-equiatomic compositions. The presence of multiple

elements in a solid solution increases the system configurational entropy, via the entropy of mixing, to overcome the unfavorable enthalpy of mixing. This entropic effect stabilizes the high-entropy solid solution relative to phase separation or formation of ordered

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intermetallic phases, resulting in enhanced mechanical properties³. As research progressed, high-entropy alloys (HEAs) were observed to provide properties and property combinations unavailable in conventional binary or ternary alloys, mainly including high-entropy effects, severe-lattice-distortion, sluggish diffusion, and cocktail effects⁴. These fascinating characteristics can not only be used for structural materials but also have found promising applications in energy and electronic applications recently^{5,6}, as components of catalysts⁷, batteries⁸, thermoelectrics⁹, superconductors¹⁰, and sensors¹¹. For example, in catalysis applications, the complex atomic configuration provides a diverse range of adsorption sites and a near-continuous binding energy distribution pattern for optimizing catalyst activity and selectivity¹². Additionally, increased configurational entropy can also offer low Gibbs energy positions for active sites, which contributes to catalyst structural stability under harsh reaction conditions¹³.

Traditional HEAs are fabricated by top-down methods like vacuum arc melting¹⁴, which produce bulk materials with limited uniformity and crystallite size control and are not suitable for catalysis and other applications where high specific surface area is needed. Recently, many advanced methods have been developed for the synthesis of HEA nanomaterials¹⁵. For example, wet-chemistry synthesis methods, such as co-reduction¹⁶, continuous-flow reduction¹⁷, Mo-assisted reduction diffusion¹⁸, and CO adsorption reduction¹⁹, can achieve low-dimensional HEA nanomaterials with small particle size or controllable morphology. However, under equilibrium reaction conditions, these methods can only integrate elements with similar physicochemical properties (such as crystal structure, atomic radius, and electronegativity) into a single-phase alloy solid solution. Phase-separation will occur when the entropy of mixing is insufficient to overcome the positive enthalpy of mixing, resulting in a limited compositional space at the atomic scale²⁰. Some special equilibrium synthesis approaches can overcome elemental immiscibility, e.g., the use of liquid gallium to decrease system mixing enthalpy^{21,22}, but generate large nanoparticles due to the high surface tension, fluidity and Ostwald ripening effect of Ga. Achieving high loadings of HEA nanoparticles is desirable for enhancing current density and reducing mass transport resistance in electrochemical devices; however, a trade-off between particle size and loading is commonly encountered²³.

In recent years, non-equilibrium synthesis approaches have been developed to fabricate HEA nanomaterials, such as flash Joule heating²⁴, high-temperature plasma²⁵, graphdiyne-assisted fast sparking platform²⁶, and laser scanning ablation²⁷. These methods provide high reaction temperature and rapid material formation on timescales of milliseconds to nanoseconds. These short timescales result in kinetic mixing of otherwise immiscible metals in a single-phase metastable alloy. However, their violent synthesis environments often lead to inhomogeneous particle size distribution, and the high-energy input and harsh operation conditions limit industrial application of these methods. Therefore, development of a versatile and scalable methodology to fabricate highly homogeneous HEA nanoparticles at high loadings on support materials is important. Likewise, understanding the mechanisms of rapid reaction kinetics and configurational entropy on particle dispersion, elemental miscibility, and metal reducibility in the non-equilibrium synthesis processes remains challenging.

Flame aerosol processing, including flame spray pyrolysis (FSP), has been one of the most common technologies to fabricate inorganic powders in industry²⁸. Benefiting from this rapid, one-step, and continuous synthesis route, millions of tons of flame-made nanoparticles like carbon black, fumed SiO₂, TiO₂, and Al₂O₃ have been manufactured by industry leaders like DuPont and Evonik^{29,30}. In an FSP process, the reactant precursors are first vaporized or atomized, and products are then formed in the vapor phase or within droplets³¹. The rapid gas-to-particle or droplet-to-particle conversion can combine

immiscible elements in a single-phase bi-metallic or high-entropy nanostructure^{13,32–34}. Traditional FSP techniques dissolve metal salts in organic solvents with high combustion enthalpy, driving particle formation through combustion of this precursor solution. The use of organic solvents and precursors soluble in them increases production costs, relative to use of inorganic salts dissolved in water, and particle formation within the flame limits the use of carbon and other thermally unstable supports (such as metal–organic frameworks, MOFs) that cannot withstand flame conditions. Flame-assisted spray pyrolysis (FASP) allows the use of aqueous precursors³⁵. Normally, in the gas-to-particle route, both FSP and FASP produce particles ranging from a few nanometers to several tens of nanometers in size, whereas the droplet-to-particle route generally yields much larger particles^{36–38}. Additionally, the flame-based candle combustion synthesis method can produce sub-100 nm carbon spheres (candle soot)³⁹, and a recent study has shown that this method can in-situ deposit several-nanometer HEA nanoparticles onto carbon nano-sphere supports⁴⁰.

In this work, we present a modified FASP reactor to produce supported HEA nanoparticles. In contrast to conventional FSP, our process separates the flame and particle formation processes into different regions of the reactor and employs a fuel-rich hydrogen flame in an enclosed system. This yields an in-situ H₂ reducing atmosphere and allows use of water as a solvent. Compared to metal salts dissolved in organic solvents, this approach not only expands the range of accessible elements but also makes the process more economically viable. Meanwhile, our specific FASP configuration enables flexible temperature control and provides much lower reaction temperature than conventional FSP and flame synthesis, which allows use of carbon and MOF materials as supports for metal nanoparticles. In this case, we produce highly dispersed HEA nanoparticles on a range of 1- to 3-dimensional support materials, including TiO₂ nanocrystals, carbon nanotubes, graphene, carbon microspheres, Zr UiO-66 MOF, Fe-N-C nanoplatelets, and simultaneously (in-situ) formed mesoporous silica. Meanwhile, we show that the rapid flame reaction kinetics and increased configurational entropy, along with heterogeneous nucleation on a support, can significantly decrease alloy particle size. We also demonstrate the role of entropy in promoting the alloying process, enabling the incorporation of enthalpically unfavorable metals. As an energy-related application, we show that RuPdOsIrPt/graphene exhibits high catalytic activity and stability for hydrogen oxidation reaction, dramatically outperforming commercial and conventional catalysts.

Results

Synthesis mechanism

The supported HEA nanoparticles were fabricated by the single-step, continuous, and in-situ reducing flame aerosol process described in the Methods section (Fig. 1a, Supplementary Figs. 1, 2). A homogeneous precursor suspension containing five different metal cations and commercial carbon black support (Supplementary Fig. 3) was continuously injected into the reactor, while the product was continuously collected downstream. In general, the non-equilibrium flame synthesis adopted here involves four mechanisms. The first is kinetically-driven mixing. When the precursor suspension entered the reactor, it was immediately shear-atomized by a sonic-velocity hot stream of combustion products into droplets of a few micrometers in diameter^{41,42}, and the evaporation time of such a small droplet was on the order of a few to several tens of milliseconds, with heating and cooling rates on the order of 10⁵–10⁶ K/s⁴³. Therefore, the HEA nanoparticle nucleation and growth occurred on a timescale too short for atoms of different elements to segregate via diffusion. This results in kinetic mixing (trapping) of different atoms in a single-phase, including immiscible elements³². A second key mechanism is entropy-driven mixing. According to $\Delta G = \Delta H - T \cdot \Delta S$, the increase in configurational entropy due to mixing decreases the system's Gibbs energy,

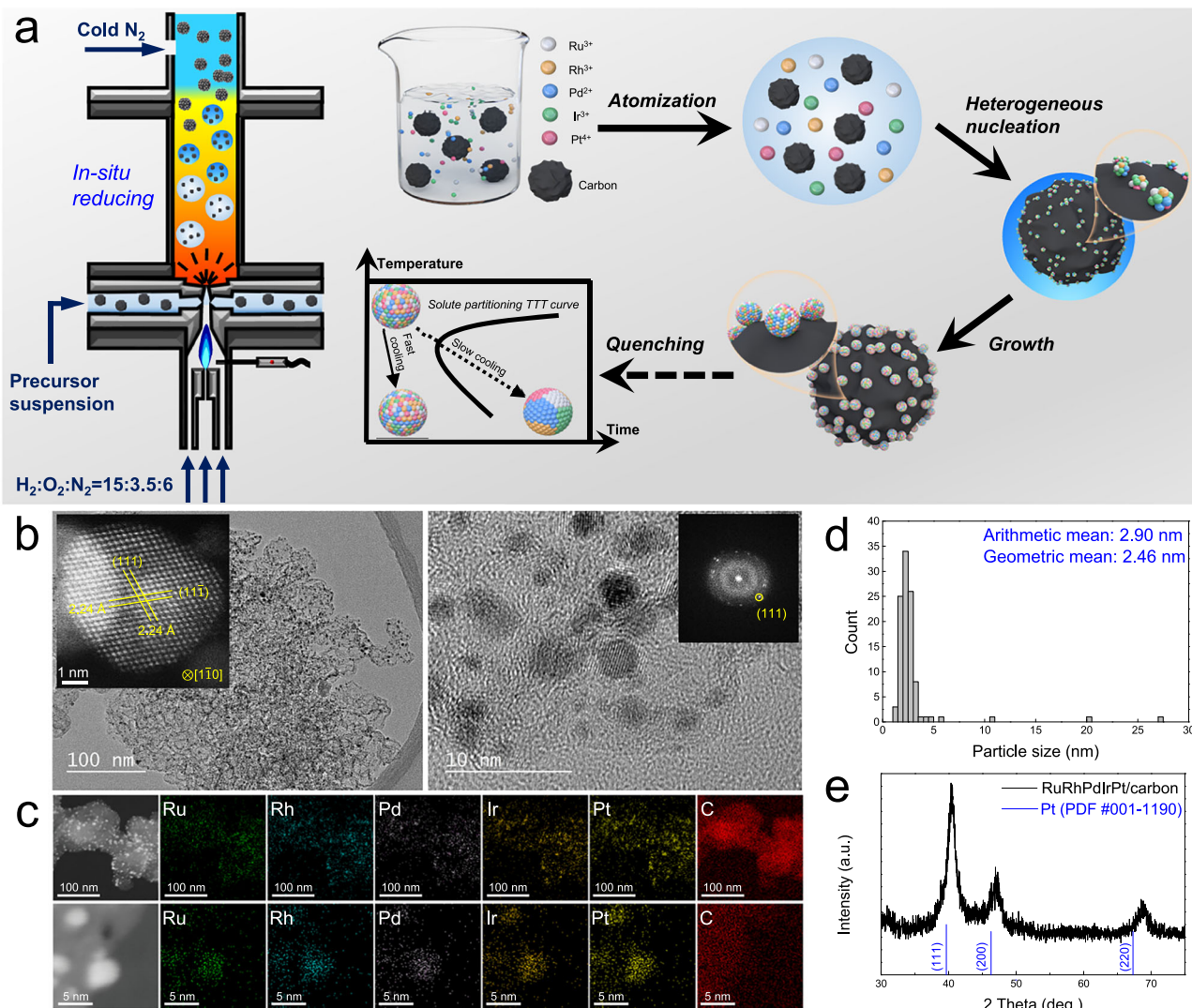


Fig. 1 | HEA nanoparticle formation process. **a** Schematic of non-equilibrium reducing flame aerosol synthesis of supported HEA nanoparticles; **b** TEM images, FFT pattern, and AC-STEM images; **c** HAADF-STEM elemental maps of RuRhPdIrPt

on commercial carbon microsphere support; **d** particle size distribution; and **e** XRD pattern of a representative RuRhPdIrPt/carbon. Source data for **d**, **e** are provided in the Source Data file.

facilitating the incorporation of multiple elements into a homogeneous solid solution⁴⁴. The third key mechanism is in-situ reduction. Although the metal precursors are introduced as ions in solution, we supplied excess H₂ to the flame, thereby creating a reducing atmosphere in the reaction chamber, which resulted in the formation of metals rather than oxides for materials that can be reduced by H₂ in the presence of H₂O. The fourth key mechanism is heterogeneous nucleation. We included a support material in the precursor suspension, which shifted the particle nucleation mechanism from homogeneous nucleation^{45,46} to heterogeneous nucleation and significantly decreased the energy barrier for nucleation. In this case, the surface of the support material provided numerous nucleation sites to disperse alloy particles, which greatly decreased the particle size from tens of nanometers, typical of a gas-to-particle formation route by traditional flame spray pyrolysis³⁴, to below 3 nm. For example, the support-free HEA formed in our current flame aerosol process exhibited large particles. This sample also exhibited phase separation between RuRhPdIrPt high-entropy alloy and (RuRhPdIrPt)O high-entropy oxide (Supplementary Fig. 4a). H₂ temperature-programmed reduction (H₂-TPR) confirms that these oxidized noble metals can be readily reduced to their metallic state at low temperatures (Supplementary Fig. 4b) and integrated into the alloy phase, forming a homogeneous HEA

(Supplementary Fig. 4c). Note that a large flow of cold N₂ was supplied at the reactor exit, to rapidly reduce the temperature, preventing particle aggregation and phase separation, thereby ensuring the formation of highly dispersed and homogeneous HEA nanoparticles.

In this manner, for example, we obtained RuRhPdIrPt/carbon with a metal:carbon loading of 40 wt.%. As shown in transmission electron microscopy (TEM) images (Fig. 1b, Supplementary Fig. 5), the HEA nanoparticles are highly dispersed on the carbon support with a geometric average diameter of 2.46 nm (Fig. 1d). The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) elemental maps of Ru, Rh, Pd, Ir, and Pt showed uniform distribution in the alloy nanoparticles with no elemental segregation (Fig. 1c). Meanwhile, the aberration corrected scanning transmission electron microscopy (AC-STEM), live fast-Fourier-transform (FFT) pattern, and X-ray diffraction (XRD) pattern confirmed the single FCC crystalline structure with no phase separation (Fig. 1b, e), demonstrating the successful mixing of all elements at an atomic level. Inductively coupled plasma optical emission spectrometry (ICP-OES) and X-ray fluorescence (XRF) analysis showed similar atomic content of the five metals with small standard deviation, in which the minor deviations are attributed to the differing vapor pressures of the metal salts²⁴, while the Energy Dispersive X-ray Spectroscopy (EDS) result

(Supplementary Fig. 6) was limited by calibration uncertainty and peak overlap, leading to larger differences from the ICP and XRF results. Compositions from all three techniques are included in Supplementary Table 1, but the EDS result cannot be considered quantitatively accurate. In addition, the Raman spectra of pure carbon support and RuRhPdIrPt/carbon were nearly identical, with similar I_D/I_G ratio, demonstrating the rapid flame process did not damage the carbon structure (Supplementary Fig. 7). N_2 ad-/desorption analysis showed the high porosity and surface area of RuRhPdIrPt/carbon with a Brunauer-Emmett-Teller (BET) specific surface area of 508 m^2/g (Supplementary Fig. 8). In another flame synthesized RuRhPdIrPt/carbon with a higher metal:carbon loading of 80 wt.% (Supplementary Fig. 9), the HEA still maintained the small geometric mean particle size of 3.80 nm. Together, these material characterization results conclusively demonstrate the successful synthesis of single-phase, highly homogeneous, and highly dispersed HEA nanoparticles.

Extension to a diverse range of support materials

To demonstrate the versatility of this method, we applied it to a diverse range of 0- to 3-dimensional support materials, including TiO_2 nanocrystals (Supplementary Fig. 10), carbon nanotubes (Supplementary Fig. 11), graphene (Supplementary Fig. 12), Zr UiO-66 MOF (Supplementary Fig. 13), and Fe-N-C prepared as reported in our previous study⁴⁷ (Supplementary Fig. 14). These support materials decorated with metal nanoparticles have broad potential for application in energy-related fields, such as photocatalysis⁴⁸, sensing⁴⁹, H_2 storage⁵⁰, organic synthesis⁵¹, and electrocatalysis⁵². Fig. 2a shows representative examples of the aerosol-synthesized 10 wt.% RuRhPdIrPt/ TiO_2 , 40 wt.% RuRhPdIrPt/Carbon nanotube, 40 wt.% RuRhPdIrPt/Graphene, 10 wt.% RuRhPdIrPt/UiO-66. The high-entropy material system is not limited to specific compositions; therefore, we also developed a 40 wt.% PdOsIrPtAu/Fe-N-C. TEM imaging revealed that HEA nanoparticles were highly dispersed across all the support materials (Fig. 2a, Supplementary Figs. 15–19), coexisting with a few anomalously larger particles. These anomalously large particles are likely caused by the uneven distribution of the support materials in the precursor suspension. The measured average diameters of HEA nanoparticles were 4.76 nm on TiO_2 nanocrystals, 2.76 nm on carbon nanotubes, 2.92 nm on graphene, 4.00 nm on UiO-66 MOF, and 4.09 nm on Fe-N-C (Fig. 2c, Supplementary Fig. 20). The HAADF-STEM element maps confirmed the homogeneous mixing of all constituent elements within each alloy nanoparticle (Fig. 2b). Meanwhile, all the XRD patterns showed a single-phase FCC crystalline structure (Fig. 2d, Supplementary Fig. 21). As observed for HEA/carbon, the Raman spectra of HEA/carbon nanotubes, HEA/graphene, and HEA/Fe-N-C were consistent with those of the pure support materials, indicating no damage to their carbon structure (Fig. 2e, Supplementary Fig. 22). All the materials retained their porosity and pore size distribution after loading with HEA nanoparticles (Fig. 2f, Supplementary Fig. 23). The observed decreases in BET surface area compared to pure support materials can be attributed to addition of mass (in the form of HEA nanoparticles) without addition of surface area, and possibly some pore blockage by the HEA nanoparticles. These results collectively demonstrate that our flame aerosol process can be successfully applied to various support materials, greatly expanding its potential applications.

Simultaneous formation of HEA nanoparticles and mesoporous silica support

Mesoporous silica is a widely used support material in heterogeneous catalysis due to its high porosity and structural stability⁵³. In a previous study⁵⁴, we presented an evaporation-driven micelle self-assembly and inorganic macromolecular sol-gel process to synthesize mesoporous silica in this flame aerosol process. It produced a mesoporous silica material with hollow nanoshell morphology, amorphous structure,

BET surface area of 856 m^2/g , and pore size distribution of 1–5 nm (Supplementary Fig. 24). Here, we found that adding metal salts to the mesoporous silica precursor solution allowed formation of mesoporous silica supported HEA nanoparticles in a single step (Fig. 3a). After atomization, the precursor droplet contained Ru, Rh, Pd, Ir, and Pt cations, along with tetraethyl orthosilicate (TEOS) and cetyltrimethylammonium bromide (CTAB). As the droplet evaporated, CTAB self-assembled into a soft template of cylindrical micelles, and TEOS molecules co-polymerized around the micelles via sol-gel chemistry, forming the intermediate of mesoporous silica. At the same time, HEA nanoparticles formed on the intermediate via the mechanisms shown in Fig. 1a. Removal of the micelle template and other organic residues (Fig. 3b) by simple ethanol washing produced the final HEA/Mesoporous silica product.

In this manner, both RuRhPdIrPt/mesoporous silica and PdOsIrPtAu/mesoporous silica materials with metal: SiO_2 loadings of 15 wt.% were successfully fabricated. Both types of HEA nanoparticles exhibited high dispersion, small and uniform particle size, and homogeneous mixing of all constituent elements within each nanoparticle at the atomic level (Fig. 3d, f, Supplementary Fig. 25). Their single FCC phase was also confirmed by XRD patterns (Fig. 3e, Supplementary Fig. 26). Even with loading of HEA nanoparticles, the materials provided high BET surface areas of 517 and 526 m^2/g for RuRhPdIrPt/silica and PdOsIrPtAu/silica, respectively, and large volumes of mesopores (Fig. 3c, Supplementary Fig. 27).

The simultaneous formation of mesoporous silica and HEA nanoparticles could greatly decrease the preparation cost of these materials relative to sequential processes in solution, improve the dispersity of the HEA nanoparticles, and prevent the formation of anomalous large particles. Additionally, the formed HEA nanoparticles were embedded within the mesoporous silica rather than forming on the surface. This nanoscale confinement can significantly enhance the sintering resistance of the nanoparticles⁵⁵.

Role of kinetics and entropy in decreasing particle size

The good dispersibility of flame aerosol synthesized HEA nanoparticles prompted us to probe this effect further. For example, in a previous study on fabricating graphene oxide supported HEA nanoparticles *via* fast moving bed pyrolysis⁵⁶, the HEA particle size was ~2 nm at a low loading of 3 wt.%. When the HEA loading was increased to 10 wt.%, the particle size grew to over 20 nm. In contrast, in our flame synthesized HEA/Carbon, the particle size of HEA increased from ~2.9 nm at 40 wt.% metal:carbon loading (Fig. 1d, Supplementary Fig. 5) to only 4.3 nm at high loading amount of 80 wt.% (Supplementary Fig. 9). Meanwhile, the mentioned study⁵⁶ also discussed the effect of HEA nucleation and growth time on the particle size. The ~2 nm HEA was achieved through a fast 5 s material formation process at 923 K. When the heating time was extended to 30 minutes, the particle size exceeded 100 nm, and phase separation occurred. However, in the current flame aerosol process, the HEA formed within a few milliseconds, as microdroplets evaporated at ~700 °C⁴³, and downstream quenching with cold N_2 prevented further growth or sintering. Therefore, we hypothesize that the non-equilibrium kinetic processes and high configurational entropy adopted here, can greatly limit the growth of metal nanoparticles, reducing the overall particle sizes (Fig. 4a). To further confirm our hypothesis, we used a traditional wet-impregnation method to load Pd nanoparticles on a typical mesoporous silica MCM-41 (ex situ loaded Pd/MCM-41) at 500 °C, which is a typical temperature in thermal annealing process for preparing catalysts⁵⁷. Meanwhile, we used the non-equilibrium flame aerosol process to load Pd nanoparticles on mesoporous silica (in-situ formed Pd/F- SiO_2) and RuRhPdIrPt HEA nanoparticles on mesoporous silica (in-situ formed HEA/F- SiO_2). The metal: SiO_2 ratio range varied from 1 wt.% to 50 wt.% and their ratio in the product is shown in Supplementary Tables 2–4.

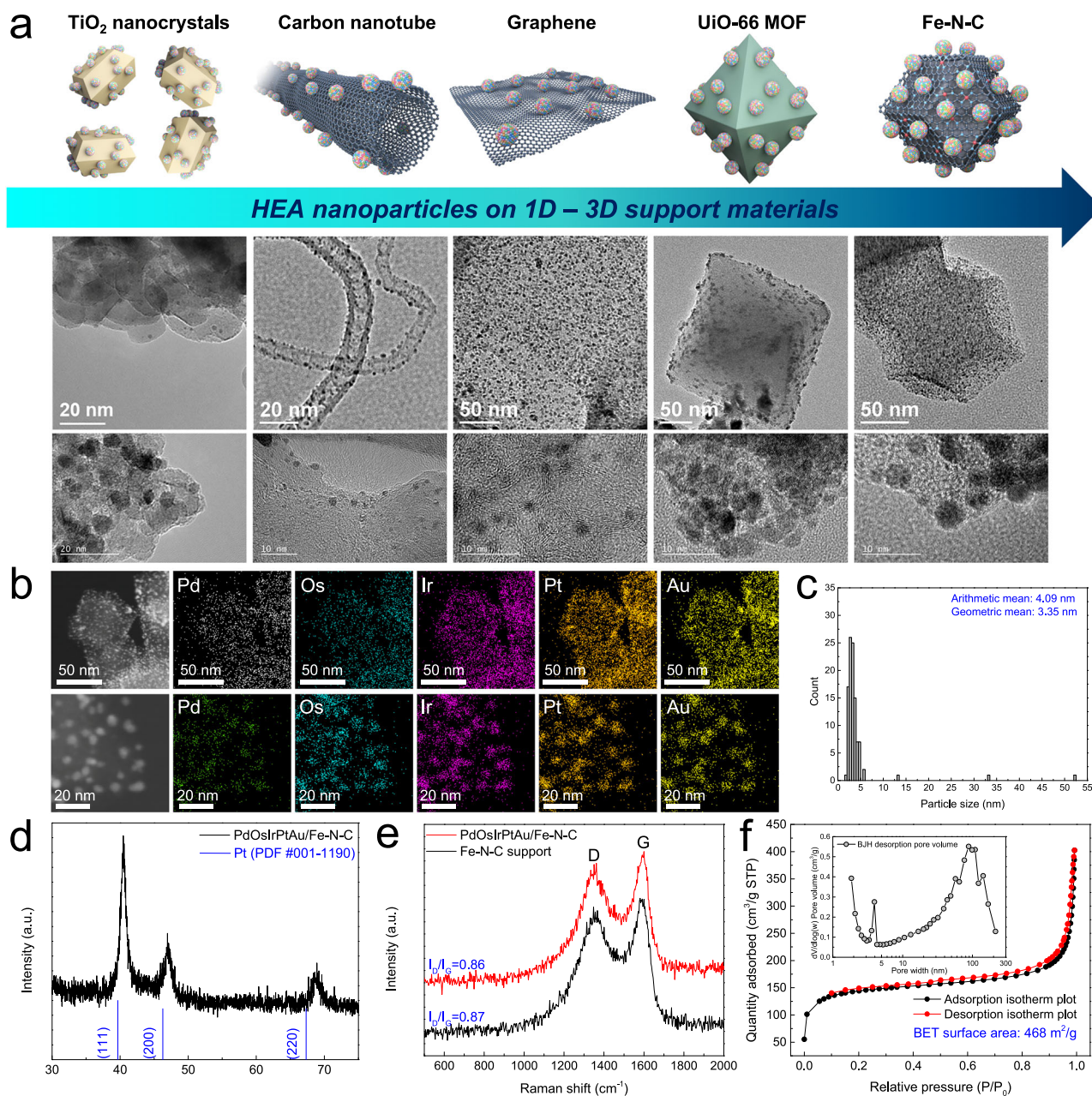


Fig. 2 | HEA nanoparticles on 1D-3D support materials. **a** Schematic and TEM images of HEA nanoparticles on various supports; **b** HAADF-STEM elemental maps of PdOsIrPtAu on Fe-N-C support; **c** particle size distribution (HEA loading 40 wt.%);

d XRD pattern; **e** Raman spectra; **f** N_2 adsorption/desorption isotherm plot and BJH desorption pore width distribution curve of PdOsIrPtAu/Fe-N-C. Source data for **c–f** are provided in the Source Data file.

The results indicated that under equilibrium reaction conditions of thermal annealing at 500 °C for Pd/MCM-41, heavy sintering of Pd nanoparticles occurred. As the Pd:SiO₂ ratio increased from 1 wt.% to 50 wt.%, the average size of Pd nanoparticles increased from 10.2 nm to 60.9 nm, and the Pd nanoparticle size distribution broadened substantially (Fig. 4b, Supplementary Fig. 28). In contrast, the non-equilibrium flame aerosol synthesized Pd/F-SiO₂ exhibited small nanoparticle size and narrow particle size distribution. As the Pd:SiO₂ ratio increased from 1 wt.% to 50 wt.%, the average size of Pd nanoparticles increased from 3.05 nm to 8.51 nm, demonstrating the role of rapid reaction kinetics in decreasing particle size (Fig. 4b, Supplementary Fig. 29). This can be attributed to their different particle nucleation and growth rates. Generally, the formation of nanocrystals undergoes the “Precursor → Monomer → Nuclei → Nanocrystal” process according to LaMer’s nucleation mechanism (Supplementary

Fig. 30)⁵⁸. The traditional co-precipitation method involved a slow heating process, leading to the gradual decreasing of metal precursor concentrations. Consequently, it failed to rapidly achieve sufficient monomer concentration for the formation of small-sized nuclei. Due to this slow heating rate and progressive metal reduction, nanoparticles have more time to adsorb monomers during independent growth, resulting in the formation of larger particles. In contrast, the non-equilibrium flame aerosol process ensured rapid thermal decomposition of precursors at high temperatures through swift heating (<50 ms), which generated a much higher degree of supersaturation and higher nucleation and growth process than equilibrium reaction (Supplementary Fig. 30). High supersaturation decreased the energy required (ΔG^*) for nucleation, thereby reducing the size of the critical nucleus (r^*) (Supplementary Fig. 31)⁵⁶. The small critical nucleus served as the foundation for nanoparticle growth, resulting in smaller

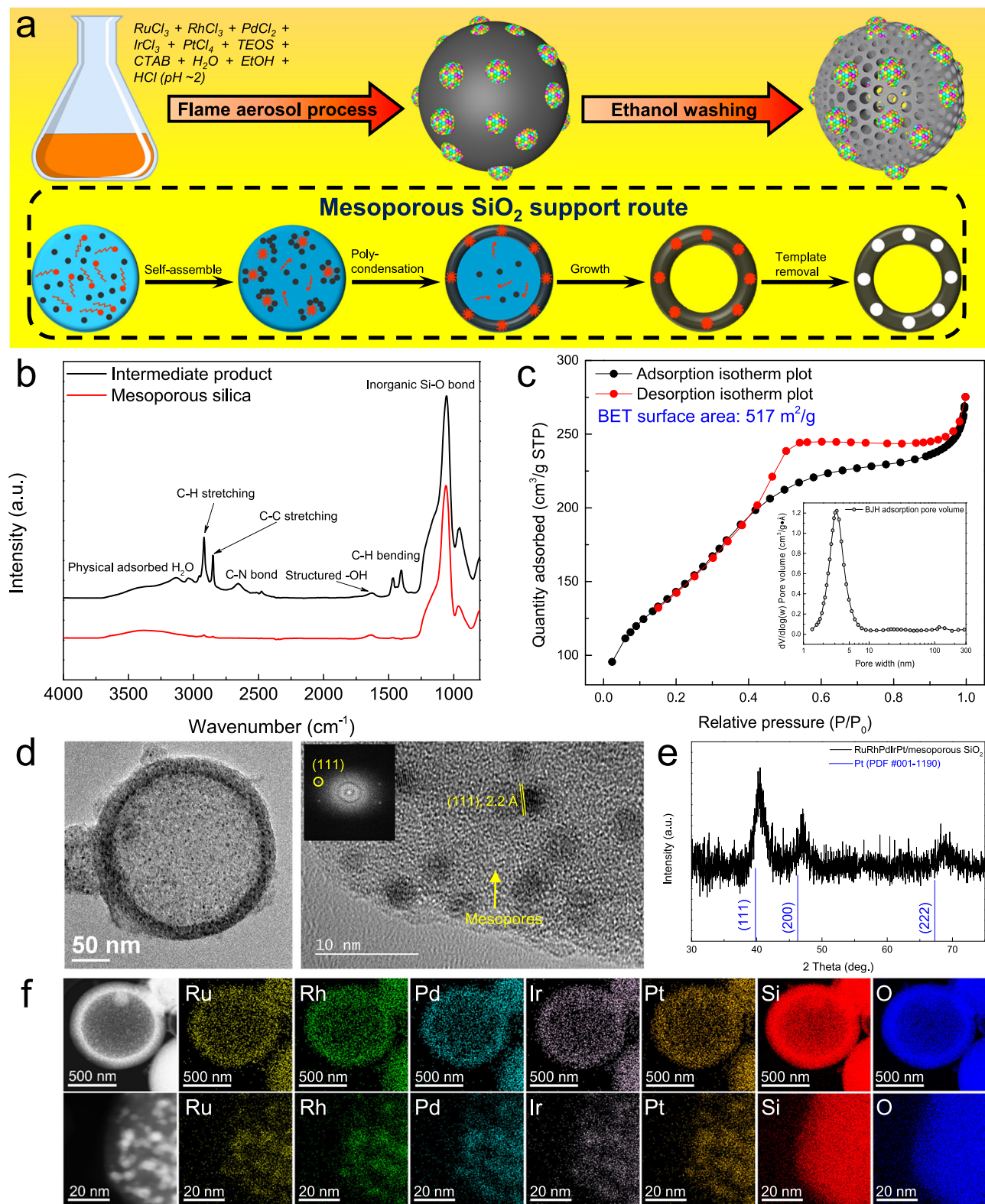


Fig. 3 | HEA nanoparticles on mesoporous silica. a Schematic of the process of simultaneous HEA nanoparticle and mesoporous silica support formation; **b** FTIR spectra of mesoporous silica before and after template removal; **c** N_2 adsorption/desorption isotherm plots and BJH adsorption pore width

distribution curve; **d** TEM images and FFT pattern; **e** XRD pattern of RuRhPdIrPt/Mesoporous silica; and **f** HAADF-STEM elemental maps of RuRhPdIrPt embedded within the mesoporous silica. Source data for **b**, **c**, **e** are provided in the Source Data file.

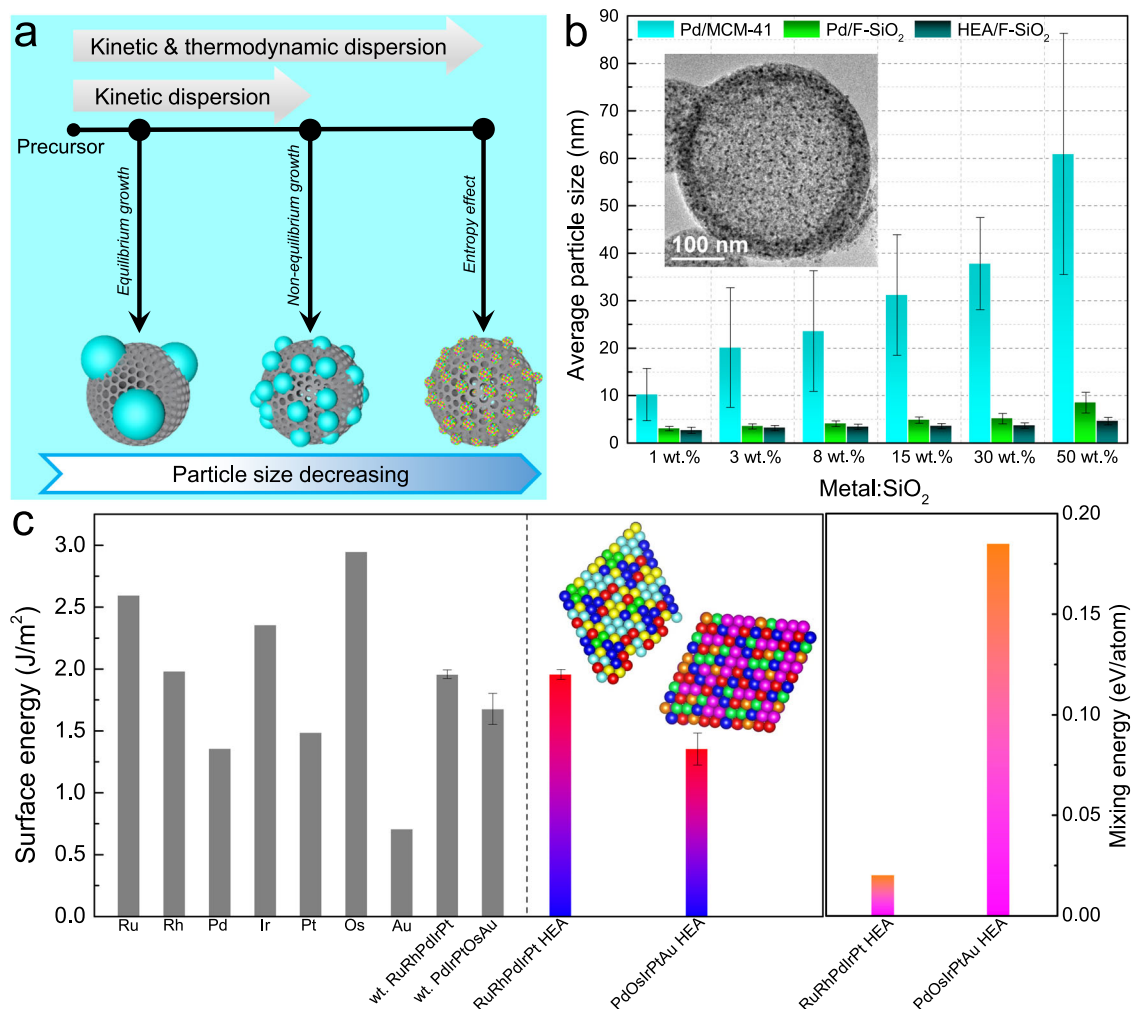


Fig. 4 | Decrease in particle size. **a** Schematic of non-equilibrium flame process and entropic effects on decreasing particle size; **b** Average particle size of wet-impregnation synthesized Pd on MCM-41 mesoporous silica (Pd/MCM-41) prepared in the liquid phase, flame-synthesized Pd on in-situ formed mesoporous silica (Pd/F-SiO₂), and flame-synthesized HEA on in-situ formed mesoporous silica (HEA/F-SiO₂), at varied metal:SiO₂ ratio of 1 - 50 wt.%. The insert TEM comes from the

50 wt.% HEA/F-SiO₂. The error bars represent the standard deviation of the particle size; **c** DFT calculation of surface energy of HEA and pure metal (111) surfaces, with the mixing energy of the HEA also provided. The error bars represent the standard deviation of the surface energy. Source data for **b**, **c** are provided in the Source Data file.

nanoparticles. Meanwhile, the fast material formation process followed by rapid quenching can dramatically limit the particle aggregation and coalescence observed in equilibrium synthesis methods, aiding in controlling the final nanoparticle size and uniformity.

In our non-equilibrium flame aerosol method, we further observed that the HEA/F-SiO₂ exhibited even smaller particle size, *i.e.*, better dispersion than Pd/F-SiO₂. As the HEA:SiO₂ ratio increased from 1 wt.% to 50 wt.%, the average size of HEA nanoparticles only increased from 2.66 nm to 4.62 nm (Fig. 4b, Supplementary Fig. 32). The 50 wt.% HEA/F-SiO₂ still retained highly dispersed nanoparticles and narrow size distribution (Supplementary Fig. 33), suggesting a role of the HEA composition in decreasing the particle size. To explain this phenomenon, first-principles calculations based on Density Functional Theory (DFT) were conducted to investigate the formation of HEA nanoparticles. For modeling equimolar quinary HEA systems, we generated a 125-atom Special Quasi-random Structure (SQS) by optimizing near-neighbor pair and many-body correlation functions through a Monte Carlo search approach⁵⁹. The SQS was fully relaxed, and surface energies for three (111) terminations were calculated using a slab model with the bulk equilibrium lattice constants held fixed (Fig. 4c, Supplementary Fig. 34, Supplementary Table 5). The critical radius for nucleation can be estimated using classical nucleation theory as

$r_{crit} = -\frac{\gamma}{\Delta G_v}$, where γ represents the surface energy, and ΔG_v is the bulk Gibbs energy change during nucleation. The PdOsIrPtAu system exhibited a low average (111) surface energy of 1.35 J/m², compared to the weighted average (111) surface energy of the constituent pure metals (1.68 J/m²). However, this system showed a high bulk enthalpy of mixing of 0.19 eV/atom. In contrast, the RuRhPdIrPt system had a similar surface energy (1.96 J/m²) to the weighted average for pure metal surfaces but displayed a much lower bulk enthalpy of mixing (0.02 eV/atom). At a synthesis temperature of 800 °C, the configurational entropy contribution for the Gibbs energy of a random equimolar quinary system was calculated as 0.14 eV/atom ($k_B T \ln n$ with $n = 5$). This high configurational entropy significantly reduces the bulk Gibbs energy and thus lowers the critical nucleation radius for both systems, increasing the nucleation rate for HEA nanoparticles relative to single-component nanoparticles. Following nucleation, classical growth theory suggests the growth of HEA nanoparticles requires the transport of constituent elements to the nanoparticle surface via a concentration gradient perpendicular to the solid-liquid interface⁶⁰. In a concentrated multi-component solution, a sufficient concentration gradient is difficult to sustain for all constituent elements, limiting the size that HEA nanoparticles can grow. Based on the Classical growth model, this condition limits the size during HEA nanoparticle growth,

resulting in similar nanoparticle sizes and narrow size distribution as observed in our experiments⁶⁰. This observation is consistent with our previous report on high-entropy ceramic oxides, where increasing the configurational entropy of the solid solution induced a transition from long-range to short-range order, thereby reducing crystallinity and grain size⁶¹.

Similarly, a recent report also described the preparation of high-density, highly dispersed HEA nanoparticles on carbon supports by a spray drying aerosol process²³. Supplementary Table 6 summarizes reports on the preparation of HEA nanoparticles using combustion or pyrolysis processes. As can be seen there, both the spray drying aerosol method and our flame aerosol method have clear advantages in HEA loading and density. The spray aerosol method has achieved finer particle sizes. Our method, however, allows for a broader range of support materials than has been demonstrated for spray drying, especially for the HEA embedded in mesoporous silica, offering advantages in stability and loading. Additionally, our method provides an in-situ reduction environment that promotes metal reduction (even with high oxidation potential) to a certain extent.

Reduction of high oxidation potential elements

Reduction-based alloying methods are often constrained by the oxidation potential of the metal elements, resulting in phase separation between metallic and oxidized elements. To this point, we have shown that the current flame aerosol process can directly reduce noble metals, including Ru, Rh, Pd, Os, Ir, Pt, Au, and Ag⁶², which have low oxidation potential in the Ellingham diagram⁶³. Meanwhile, a recent study suggests that through thermodynamic design, the mixture of metal oxides can be directly reduced into bulk alloys by H₂ in one step⁶⁴. Here, in the nanoscale alloying process, we address the role of RuRhPdIrPt HEA in the reduction and incorporation of metals with moderate to strongly negative redox potentials that otherwise would be challenging to produce as metals in conventional reduction-based alloying methods (Fig. 5a). The entropic stabilization effect favors this reduction⁴, where the increase in system configurational entropy promotes the alloying of multiple elements in a single FCC metallic phase, along with a catalytic role of the HEA nanoparticles for breaking M-O bonds at high temperatures.

First, to theoretically investigate the catalytic reactivity of the HEA, we performed DFT calculations of oxygen desorption energies as an indicator of M-O bond strength (Fig. 5b). Here, the M-O bond is between oxygen and a metal with moderate to strongly negative oxidation potential. We considered a FCC (111) surface of the RuRhPdIrPt HEA with full coverage of oxygen in the FCC “hollow” sites⁶⁵, and substituted one of the surface atoms with a *M* atom (*M* = W, Ce, or Zr). We sampled various metal sites for *M* substitution as well as different oxygen atoms adjacent to those sites for desorption, yielding distributions of calculated desorption energies. We also calculated analogous oxygen desorption energies for surfaces of pure *M* elemental metals, i.e., W in the BCC structure, Ce in the FCC structure, and Zr in the HCP structure, by placing oxygen atoms in FCC hollow sites for Zr/Ce and “three-fold” sites for W⁶⁶. The DFT results are shown in Supplementary Table 7 and plotted in Fig. 5c for W, Fig. 5d for Ce, and Supplementary Fig. 35 for Zr. The total oxygen desorption energies for the HEA surfaces were consistently lower than those of pure metals. Oxygen desorption energies for ZrO₂, WO₃, and CeO₂ were also obtained from calculations in the literature^{67–69} and used to estimate M-O bond strength in a similar manner (Supplementary Tables 7, 8). With the exception of two cases in CeO₂, these bond energies are significantly higher than our calculated values for those metals in the HEA surface, which confirms that the HEA weakens the M-O bonds relative to the oxide environments as well.

While the total oxygen desorption energies are indeed lowest in the HEA case, this does not necessarily indicate a weakening of the M-O bonds (*M* = W, Ce, Zr), because oxygen desorption requires breaking

multiple M-O bonds in pure *M* metals, while there is only one M-O bond in the HEA case. For a better comparison, we normalized the oxygen desorption energy per M-O bond. In the pure metal and oxide cases, this is simply the total desorption energy divided by the coordination number of the oxygen atom removed (e.g., three for the pure metal surfaces). In the HEA case, however, oxygen removal involves breaking bonds between different metal species (one *M* and two non-*M* constituting the HEA). To extract M-O bond energies in this case, we fit a simple bond-breaking model in which the total desorption energy for a given oxygen site is the sum of bond energies of the three neighboring metal sites (Supplementary Table 9) and plotted in Fig. 5c, d and Supplementary Fig. 35. Based on these values, we found that the HEA environment weakens the M-O bonds by 0.5–0.7 eV relative to the pure *M* case.

In addition, we adopted another DFT calculation to investigate the Gibbs free energies for the reduction of W/Zr/Ce oxide clusters in vacuum and HEA surface cases. The results showed that ΔG of the reduction reactions changing from 0.15 eV under vacuum condition to -0.5 ± 0.3 eV on HEA (111) surface for W, from 2.38 eV to -1.0 ± 0.3 eV for Ce, and from 1.76 eV to -1.3 ± 0.3 eV for Zr (Supplementary Table 10), confirming the effect of HEA in promoting metal reduction.

We also performed experiments to incorporate other elements in RuRhPdIrPt HEA. Thermodynamically, according to $\Delta G = \Delta H - T \cdot \Delta S$, the increase in configurational entropy decreases the system's Gibbs energy, facilitating the mixing of multiple elements in the FCC metallic phase. Thus, moderate oxidation potential metal elements can be directly reduced with the help of HEA and stabilized in metallic alloys (Fig. 5a). Taking W as an example, the flame aerosol process could not reduce pure W and instead formed WO₃ oxide on the carbon support when W was the only metal in the precursor (Supplementary Fig. 36). The RuRhPdIrPtW/carbon showed a single-phase metallic FCC structure without any WO₃ phase detected (Supplementary Fig. 37). The metallic W⁰ specie was also detected from XPS analysis, demonstrating the W was successfully reduced (Supplementary Fig. 38)⁷⁰. The higher oxidation state of W may originate from post-synthesis inevitable surface oxidation in air. However, it is also possible that not all of the introduced W is fully reduced, and WO₃ clusters may be present.

This reduction mechanism can be extended to alloy other metal elements with similar or lower oxidation potential as W in the Ellingham diagram, such as Fe, Co, Ni, Cu, Zn (Supplementary Figs. 42–44). In addition, although this is not the main focus of this work, we note that in the RuRhPdIrPtW/carbon material, we observed many other structures besides HEA nanoparticles, including single-atoms, nanoclusters, and amorphous metals (Fig. 5e, Supplementary Fig. 39). A decrease in the crystallinity of the RuRhPdIrPtW HEA was also evident (Supplementary Fig. 37), but without a significant change in lattice parameters. We attribute this phenomenon to the physicochemical property differences between W and other elements (Supplementary Table 11). For the nanoparticles observed in RuRhPdIrPtW HEA samples, the elemental maps of W, like the other elements, showed homogeneous distribution throughout the particle, indicating W was successfully incorporated into the alloy lattice (Supplementary Fig. 40). However, the FCC lattice of RuRhPdIrPtW HEA nanoparticles (Fig. 5e, Supplementary Fig. 39) showed significant lattice distortion compared to RuRhPdIrPt (Fig. 1b). Specifically, we captured a particle with a significant number of defects (Supplementary Fig. 40). This particle contains a large internal cavity and is likely in an intermediate state between a nanoparticle and a nanocluster, on the verge of splitting. This provides further evidence that elements with strong immiscibility with other components (such as W) play a role in limiting alloy nanoparticle growth. Also, in a flame-synthesized binary PtW/carbon sample, in addition to highly distorted nanoparticles, numerous single atoms and nanoclusters were observed (Supplementary Fig. 41). This further demonstrates that under non-equilibrium reaction conditions, immiscible elements play a key role in disrupting the

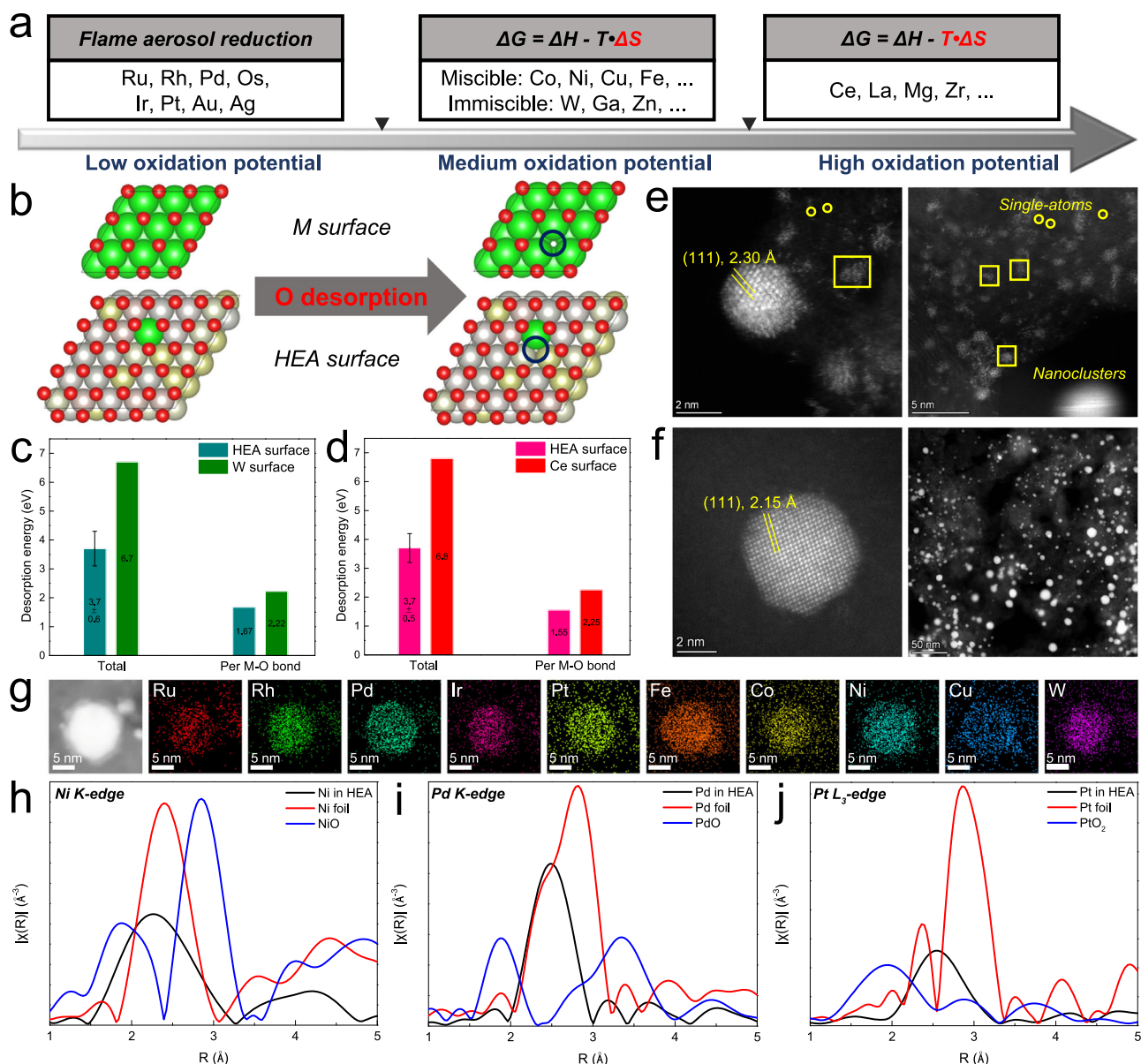


Fig. 5 | Extension of HEA compositional space. a Schematic of entropy-driven incorporation of metals with increased oxidation potential; **b** DFT calculation of O atom desorption energy; **c** O desorption energy from HEA and W surface; **d** O desorption energy from HEA and Ce surface. The error bars represent the standard deviation of the desorption energy; **e** AC-STEM images of RuRhPdIrPtW/carbon

without H₂ heating post-treatment; **f** AC-STEM images of RuRhPdIrPtCe/carbon after post-treatment in H₂ at 900 °C for 6 h; **g** HAADF-STEM elemental maps of a representative 10-element HEA/carbon; **h** Ni K-edge EXAFS spectra, **i** Pd K-edge EXAFS spectra, and **j** Pt L₃-edge EXAFS spectra of the 10-elements HEA/carbon. Source data for **c**, **d**, **h**–**j** are provided in the Source Data file.

alloy lattice and promoting the formation of more highly dispersed species. A possible reason is that, during the non-equilibrium process, rapid particle formation prevents immiscible elements from diffusing slowly to achieve phase separation to form highly crystalline, thermodynamically stable nanoparticles. Due to significant differences in atomic radius and electronegativity, the immiscible component disrupts the orderly arrangement of metal atoms, inhibiting lattice formation, causing lattice distortion and defects, and promoting the generation of more dispersed species, such as nanoclusters, amorphous particles, or even single atoms. We note that, in many reactions, these highly dispersed active sites have shown high catalytic activity⁷¹.

Specifically, using low-melting-point metals (such as Ga) is an effective approach to reducing the system's mixing enthalpy and facilitating HEA formation. However, during the HEA formation process, the surrounding metal elements tend to coalesce into Ga, leading to the formation of large particles (20–100 nm)²¹. In contrast, the non-

equilibrium method employed in this study suppresses the diffusion process. As an immiscible element, low-melting-point Ga not only avoids particle growth but, similar to W, it plays a role in inhibiting particle growth and even reducing particle size (Supplementary Fig. 45).

Furthermore, the flame aerosol process failed to directly reduce ceramic component elements with much lower oxidation potential in the Ellingham diagram, such as Ce, La, Mg, and Zr. However, according to $\Delta G = \Delta H - T \cdot \Delta S$, the increased value of $T \cdot \Delta S$ at elevated temperatures could further decrease Gibbs energy, so we adopted an H₂ heating post-treatment (800–1000 °C) to reduce and incorporate ceramic elements in HEA nanoparticles (Fig. 5a). Taking Ce as an example, the flame-synthesized RuRhPdIrPt-CeO₂/carbon was heated in H₂ at 900 °C for 6 h. The post-reduced RuRhPdIrPtCe/carbon adopted a single-phase metallic FCC structure (Supplementary Fig. 46), and the FCC crystal lattice was further confirmed by AC-STEM

(Fig. 5f). The slight deviations in lattice parameters from RuRhPdIrPt HEA demonstrated the successful incorporation of Ce. Also, the high-resolution elemental maps showed the homogeneous mixing of Ce with other constituent elements (Supplementary Fig. 47). In the Ce 3d XPS spectra of RuRhPdIrPtCe/carbon (Supplementary Fig. 48), we observed spin doublets at 904.5 eV and 886.1 eV with ΔE between the main peaks of 18.4 eV, indicating that Ce was mainly in the metallic state⁷². We attribute the slight presence of oxidized states of Ce and Pt to post-oxidation upon exposure to air. After long-term thermal annealing at 900 °C in H₂, some sintering of HEA nanoparticles occurred but numerous small nanoparticles remained, as shown in TEM images, resulting in a broader distribution with a geometric mean diameter of 6.94 nm (Fig. 5f, Supplementary Fig. 49). This mechanism can be extended to reduce and incorporate other ceramic elements (Supplementary Figs. 50–52). However, although the presence of HEA can promote the reduction and incorporation of these high oxidation potential elements, the formation of residual oxide clusters is inevitable. A ZrO₂ impurity was observed in RuRhPdIrPtZr/carbon due to the highly negative redox potential of ZrO₂. Additionally, Pt-based alloys typically transform to intermetallics under high-temperature hydrogen reduction⁷². However, no intermetallic formation was observed in any of the HEA materials studied here. We attribute this to the entropic stabilization effect of high-entropy materials⁴. In multi-component systems (≥ 4 elements), entropy dominates over enthalpy, favoring the formation of disordered alloy solid solutions over ordered intermetallic compounds⁷³.

The entropy-induced reduction mechanism enabled mixing more elements in the HEA nanoparticles. For example, a 10-element RuRhPdIrPtFeCoNiCuW/carbon was successfully fabricated (Supplementary Fig. 53). The HAADF-STEM element maps revealed the homogeneous mixing of all 10 constituent elements within the nanoparticles (Fig. 5g). The bonding structures of the 10-element HEA was measured using X-ray absorption spectroscopy (XAS), in which the Ni, Pd, and Pt elements from the 3rd, 4th, and 5th periods were selected for investigation. In their X-ray absorption near edge structure (XANES) spectra, the significantly lower white-line intensity compared to their oxides confirmed the high occupancy of their *d* orbitals (Supplementary Figs. 54–56), and the Fourier-transformed extended X-ray absorption fine structure (EXAFS) (Fig. 5h–j) spectra indicated their metallic states. In a configurationally disordered HEA, each atom is surrounded by different elements, leading to electronic interactions such as charge transfer and orbital hybridization, which modify atomic bonding. This mixed environment averages bond lengths across elements and causes deviations from their pure metal counterparts⁷⁴. All the elements showed a similar metallic bonding distance of 2.3–2.5 Å, illustrating the uniform alloy structure. In fact, the M–M bond peak positions exhibited obvious deviation compared to the corresponding metal foil references, indicating that atoms of the same element were not aggregated in the HEA nanoparticles, further supporting their homogeneous dispersion at atomic level. Meanwhile, the XPS analysis confirmed the metallic state of constituent elements in the 10-element HEA (Supplementary Figs. 57, 58). Therefore, the proposed entropy-induced reduction mechanism dramatically expands the compositional space of possible alloys. For example, by altering the composition and ratio of 10 specific metals with the ratio of each element ranging from 5 to 50%, with a 5% step size, more than 7 million different HEAs could be created⁷⁵. The non-equilibrium flame aerosol process provides a broad range of diversity for materials design and optimization, particularly if it can be coupled with machine learning strategies for navigating this enormous compositional space.

Catalysis application

In recent years, HEA nanomaterials have shown significant advantages in high-performance catalyst design⁷⁶. Their compositional diversity enables fine-tuning of catalytic activity and selectivity, and the

enhanced configurational entropy in alloy nanoparticles provides structural stability in acid, alkaline, or high-temperature reaction conditions^{6,7,12}. For example, alkaline hydrogen oxidation reaction (HOR) is one critical half-cell reaction for anion exchange membrane fuel cell applications. Unfortunately, the HOR in alkaline electrolyte suffers from sluggish kinetics due to strong hydrogen binding energy (HBE) and the participation of multiple reaction intermediates, i.e. *H, *OH and *H₂O^{77,78}. Introducing oxyphilic elements in Pt, such as RuPt, has been identified as an efficient approach to accelerate the oxidation of *H and promote the reaction kinetics^{79,80}. Further engineering the catalyst structure with multiple adsorption will be promising for further enhancing the HOR activity. Here, as an example, the flame aerosol synthesized RuPdOsIrPt/graphene high-entropy nanoparticles were employed as a hydrogen oxidation reaction (HOR)⁸¹ catalyst in an alkaline electrolyte (Fig. 6a). The optimized high-entropy catalyst showed single-phase FCC crystalline, highly dispersed HEA nanoparticles with average diameter of 2.48 ± 0.51 nm, and composition molar ratio of Ru:Pt:Os:Ir:Pt=1.09:1.48:1.00:1.50:1.93 (Supplementary Fig. 59, Supplementary Table 12). Commercial RuPt/carbon (TKK) catalyst (Supplementary Fig. 60) and graphene supported monometallic metal catalysts synthesized at the same conditions were used as references.

Cyclic voltammogram (CV) tests were first performed to investigate the electrochemical surface areas (ECSAs) of these two catalysts by the underpotential deposition of hydrogen (H_{upd}) peaks (Supplementary Fig. 61). The RuPdOsIrPt/graphene displays an ECSA of $28.3 \text{ m}^2 \text{ g}_{\text{PGM}}^{-1}$, slightly lower than that of RuPt/C catalyst ($39 \text{ m}^2 \text{ g}_{\text{PGM}}^{-1}$). The HOR activity of the HEA catalyst was assessed using a rotating disk electrode in H₂-saturated 0.1 M KOH electrolyte. Figure 6b and Supplementary Fig. 62 show the HOR polarization curves of the studied catalysts at a low loading of $10 \text{ } \mu\text{g}_{\text{PGM}} \text{ cm}^{-2}$. The HEA catalyst exhibited the lowest half-wave potential ($E_{1/2}$) of 0.014 V vs. RHE, lower than those of commercial RuPt/Carbon (0.018 V), Pt/graphene (0.031 V) and Ir/graphene (0.055 V). The exchange current densities (j_0) of the studied catalysts were further calculated by the Tafel plot derived from Butler–Volmer fitting and linear fitting of the micro-polarization region from -4 to 4 mV (Fig. 6c, Supplementary Fig. 63). The HEA catalyst demonstrated a high exchange current density of $2.67 \pm 0.16 \text{ mA cm}^{-2}$, 14% higher than commercial RuPt/Carbon (Fig. 6d), suggesting faster HOR kinetics on the HEA catalyst. In addition, we further calculated the kinetic mass activity at an overpotential of 50 mV (Fig. 6e). The HEA catalyst reveals the highest mass activity of $4.3 \pm 0.15 \text{ A mg}_{\text{PGM}}^{-1}$, which is about 7.2, 30, and 2.1 times higher than those of Pt/graphene ($0.65 \text{ A mg}_{\text{PGM}}^{-1}$), Ir/graphene ($0.15 \text{ A mg}_{\text{PGM}}^{-1}$), and commercial RuPt/carbon ($2.05 \text{ A mg}_{\text{PGM}}^{-1}$), respectively.

Moreover, we studied the stability of the HEA catalyst and commercial RuPt/carbon catalyst by accelerated stress tests (ASTs, cycling between -0.1 to 0.4 V for 10000 cycles) (Fig. 6e). The RuPdOsIrPt/graphene shows good stability. After 10000 AST cycles, the ECSA loss was only 13% (from 28.3 to $24.7 \text{ m}^2 \text{ g}_{\text{PGM}}^{-1}$, Supplementary Fig. 64a). In contrast, the commercial RuPt/Carbon lost 48% of the ECSA (from 39 to $20.4 \text{ m}^2 \text{ g}_{\text{PGM}}^{-1}$, Supplementary Fig. 64b). The HEA catalyst showed negligible changes in the HOR polarization curve. After 10000 cycles, only 1.5 mV loss in $E_{1/2}$ was observed, demonstrating the stability of the HEA catalyst. In contrast, the commercial RuPt/Carbon showed obvious activity decay after 10,000 cycles (Fig. 6f). To be more quantitative, we calculated the change in exchange current density and mass activity before and after the stability test. The HEA catalyst retained 91% and 81% of the exchange current density and mass activity (Fig. 6g), respectively, outperforming the commercial RuPt/Carbon (46% in exchange current density and 14% in mass activity).

XAS and DFT calculations were performed to further elucidate the possible origins of the enhanced HOR activity and stability of the RuPdOsIrPt/graphene catalyst. The local structures of both RuPdOsIrPt and commercial RuPt catalysts were investigated using EXAFS

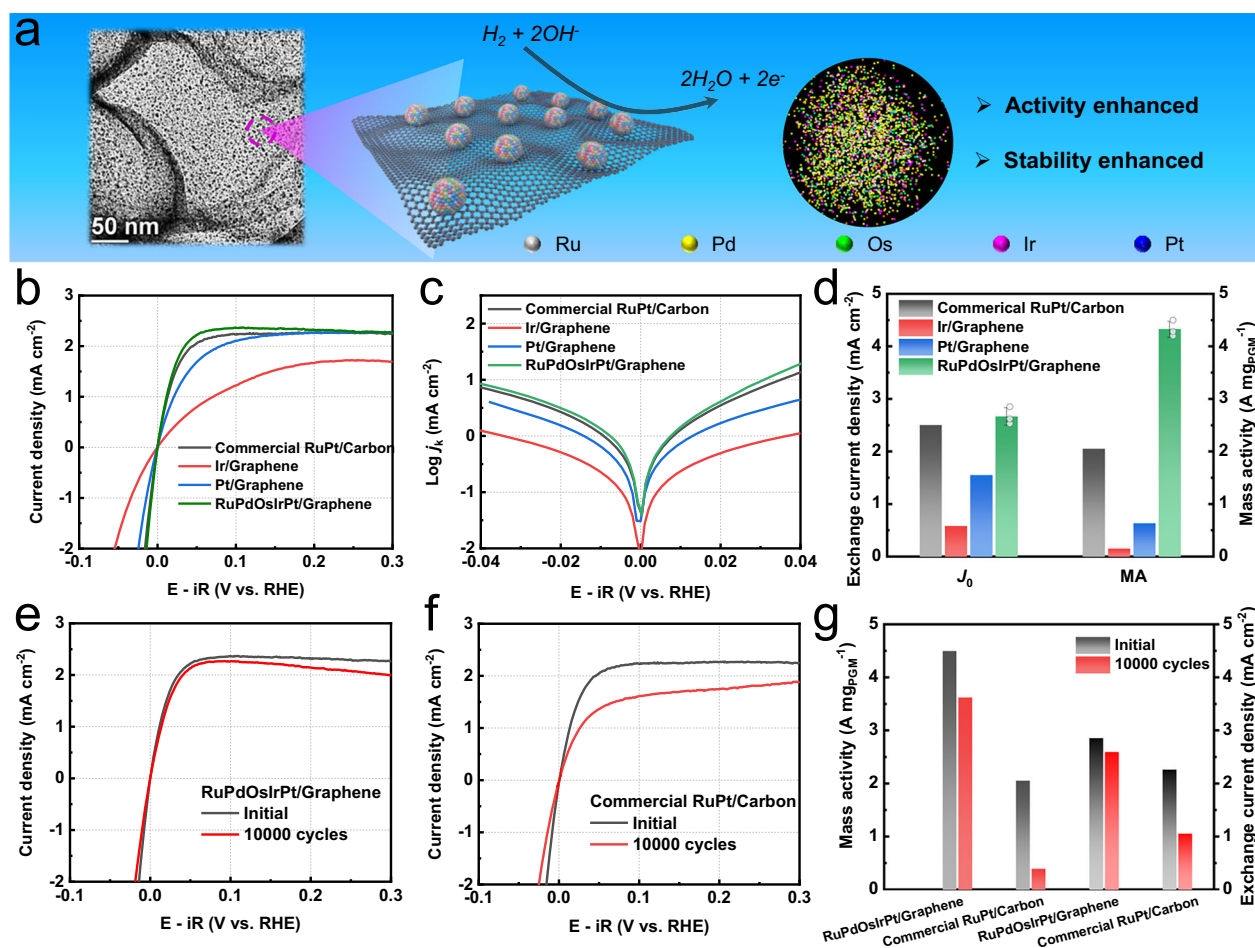


Fig. 6 | Catalytic application in hydrogen oxidation reaction (HOR).

a Illustration of the advantages of RuPdOsIrPt/graphene high-entropy catalyst; **b** 98% iR-corrected polarization curves (electrode area = 0.196 cm²); **c** Tafel plots; and **d** normalized mass activity and exchange current density of the studied catalysts; **e** HOR polarization curves of RuPdOsIrPt/graphene high-entropy catalyst before and after AST; **f** HOR polarization curves of commercial RuPt/Carbon

catalyst before and after AST; and **g** Normalized mass activity and exchange current density before and after 10000 cycles. The resistance in 0.1 M KOH was measured to be $-46.3 \pm 2.4 \Omega$. The pH is about 12.9 ± 0.1 . The error bars are the standard deviation from three separate measurements. Source data for b-g are provided in the Source Data file.

(Fig. 7a and Supplementary Fig. 65). The main peaks at $\sim 2.5 \text{ \AA}$ are assigned to metallic Ru-Ru or Ru-M scattering. Notably, in the RuPdOsIrPt catalyst, these Ru-M peaks appeared significantly broader and partially split compared to those in the commercial RuPt, indicating pronounced local lattice deformation induced by the high-entropy configuration. In high-entropy systems, such lattice distortions arise from size and electronic mismatches among constituent elements, manifesting as uneven coordination environments, strain-induced shifts in the *d*-band center, and broken local symmetry. These structural irregularities generate a diverse distribution of local electronic states and multiple non-equivalent metal-metal bond lengths, resulting in a quasi-continuous adsorption energy landscape. This, in turn, promotes the co-adsorption and activation of key HOR intermediates ($\ast\text{H}$, $\ast\text{OH}$, and $\ast\text{H}_2\text{O}$) on the catalyst surface^{82,83}. Furthermore, symmetry breaking introduces a greater variety of interfacial sites (e.g., Ru-Os, Pt-Ir), thereby increasing the density of catalytically active centers and contributing to the high performance of the HEA catalyst under alkaline conditions^{84,85}. Specifically, the Ru-M bond lengths of the RuPdOsIrPt catalyst became longer (2.689–2.701 Å) than those of the RuPt catalyst (2.668–2.699 Å, Fig. 7b, Supplementary Table 13). In general, elongated metal-metal bonds reduce orbital overlap, leading to a narrower *d*-band width and an upshift of the *d*-band center. This electronic modulation enhances the adsorption strength of reaction

intermediates⁸⁶. Correspondingly, the RuPdOsIrPt catalyst shows a higher Ru-O coordinate number ($\text{CN}_{\text{Ru-O}}$) of 1.508 compared to RuPt/C (~ 1.026), suggesting a stronger affinity of Ru sites for oxygen species. This increased oxygen affinity implies a higher hydroxyl binding energy (OHBE), which can facilitate the rate-determining step of alkaline HOR. The stronger OHBE was further supported by CO stripping experiments (Supplementary Fig. 66). The commercial RuPt/C catalyst revealed a CO stripping peak at -0.6 V , consistent with a previous report⁸⁷. In contrast, the RuPdOsIrPt catalyst showed a 50 mV negative shift in the CO stripping peak, indicating more favorable $\ast\text{OH}$ adsorption and faster CO oxidation kinetics. These findings are in good agreement with the EXAFS results, confirming that the enhanced oxygen affinity of the HEA surface plays a key role in accelerating HOR kinetics. Similarly, the Pt-M bond in the HEA was longer than that in the RuPt catalyst, leading to a stronger HBE and facilitating the dissociation of the H-H bond in H_2 (Supplementary Fig. 67).

The electronic structure of the RuPdOsIrPt catalyst was analyzed using projected partial density of states (PDOS) analysis (Fig. 7c). Notably, significant overlaps between *d*-orbitals of different metal elements were observed, indicating their enhanced electron delocalization and their strong bonding. Specifically, the Pd 4*d* and Pt 5*d* states are primarily located below the Fermi level, indicating their roles as electron-donor sites favorable for $\ast\text{H}$ adsorption. In contrast, Ru, Os,

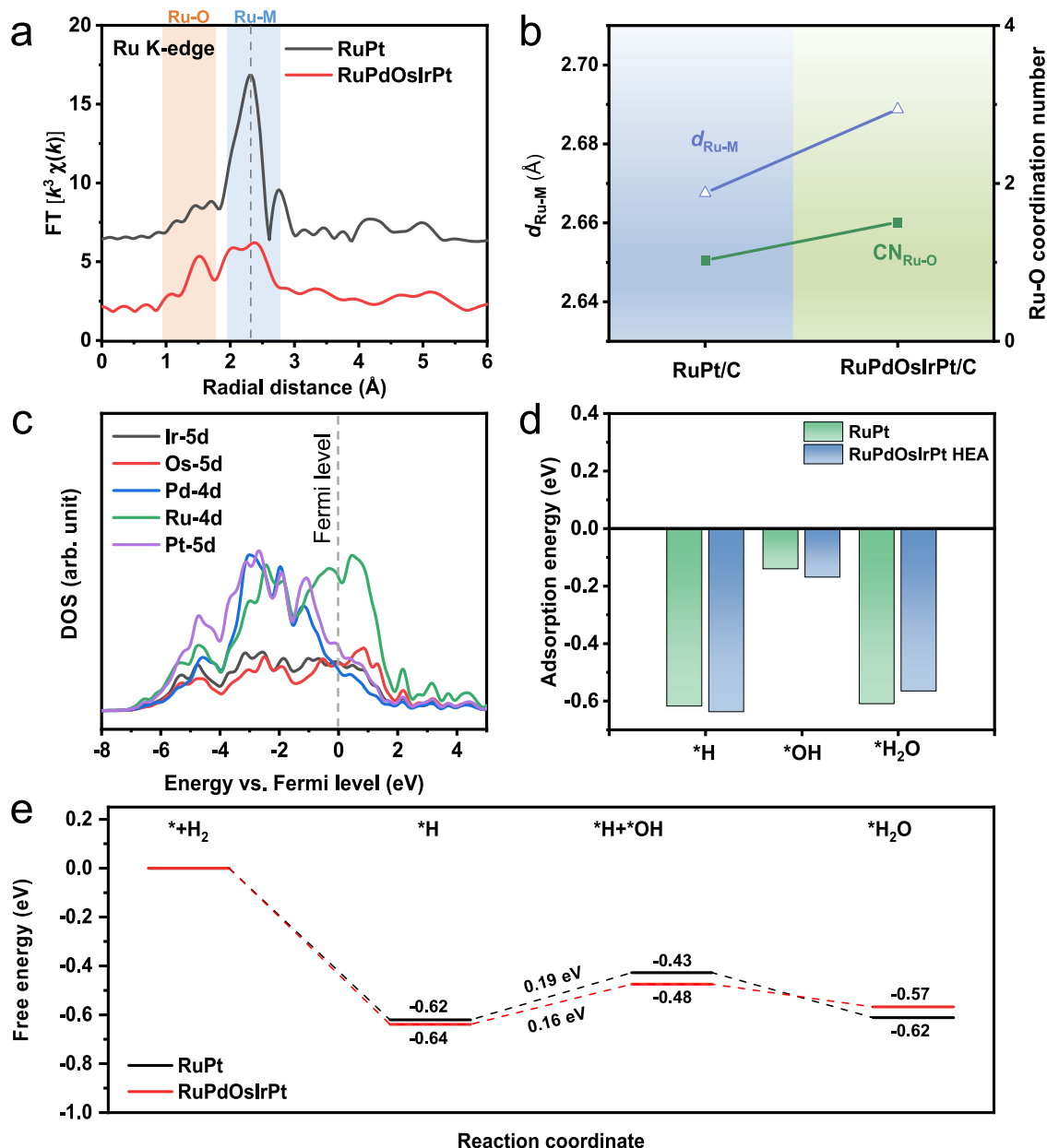


Fig. 7 | Catalysis mechanism analysis. **a** k^3 -weight FT-EXAFS spectra of the RuPt and RuPdOsIrPt catalysts in R -space. **b** Ru-M bond length and Ru-O coordination number of RuPt and RuPdOsIrPt catalysts. Computational results of **c** Site-dependent PDOS of the RuPdOsIrPt HEA model. **d** Adsorption energies of $^*\text{H}$, $^*\text{OH}$

and $^*\text{H}_2\text{O}$ on RuPt or RuPdOsIrPt surface. **e** Free energy diagrams of the HOR pathway on RuPt or RuPdOsIrPt surfaces. Source data for a-e are provided in the Source Data file.

and Ir exhibit broader band structures with substantial DOS above the Fermi level, serving as electron-acceptor sites conducive to $^*\text{OH}$ adsorption. These features imply a rich diversity of adsorption sites for accommodating various HOR intermediates. Indeed, we found that the binding energy of the intermediates on a particular site was significantly affected by the surrounding atoms. The HBE was weakened around Ir-Pt-Os sites, while the OHBE was strengthened around Ir-Ru sites (Supplementary Fig. 68). These results could demonstrate that the HEA can provide diverse adsorption sites for multiple reaction intermediates.

To further understand the catalytic behavior, we calculated binding energies of key HOR intermediates, *i.e.*, $^*\text{H}$, $^*\text{OH}$ and $^*\text{H}_2\text{O}$, based on the most stable surface adsorption sites (Fig. 7d). The RuPdOsIrPt HEA exhibited slightly stronger HBE than RuPt, which is beneficial for facilitating H_2 dissociation and consistent with the EXAFS

results. More importantly, the HEA displayed a stronger OHBE and weaker H_2O binding energy (H_2OBE) compared to RuPt. The stronger $^*\text{OH}$ adsorption promotes the removal of surface-bound protons, while the weaker $^*\text{H}_2\text{O}$ adsorption facilitates its desorption, both of which are crucial for accelerating the alkaline HOR. Additionally, we calculated reaction free energy diagrams for HOR on both RuPt and RuPdOsIrPt surfaces (Fig. 7e). The results identify the adsorption of $^*\text{OH}$ would be the (rate-determining step) RDS for HOR. Thanks to the enhanced OHBE and weakened H_2OBE on the RuPdOsIrPt surfaces, the adsorption of $^*\text{OH}$ and formation of water becomes more favorable. The RuPdOsIrPt surface achieves a low RDS energy barrier of 0.16 eV, lower than RuPt surface (0.19 eV), further confirming the high HOR activity of HEA to PtRu. Taken together, the combination of diverse active sites, tailored electronic structure, and optimized intermediate binding energies could explain the high HOR

activity of the RuPdOsIrPt HEA catalyst relative to the conventional PtRu system.

Moreover, after HOR operation, although XPS analysis did not reveal significant changes in the oxidation states of Ru and Pt in either catalyst (Supplementary Figs. 69, 70), the decreased stability of the commercial RuPt was associated with substantial particle agglomeration and sintering (evident in TEM imaging, Supplementary Figs. 71, 72) and metal dissolution (as confirmed by XRF in Supplementary Fig. 73). These degradative changes are also consistent with the observed decline in ECSA, showing a 48% loss for the RuPt catalyst compared to only a 13% loss for the RuPdOsIrPt HEA catalyst. This enhanced structural robustness in the HEA catalyst aligns with our previous study in high-entropy oxides¹³, and can be attributed to the entropy stabilization effect⁴. This effect suppresses phase segregation and reduces atomic diffusivities, thereby slowing or preventing aggregation and sintering.

Discussion

This research establishes a versatile flame aerosol methodology to produce supported HEA nanoparticles. The non-equilibrium reaction kinetics and entropy-driven alloying provide flexibility in both support materials and HEA composition, enabling rational material design across an enormous composition space for target applications. The produced HEA nanomaterials demonstrate many fascinating characteristics, such as high nanoparticle dispersion, atomic homogeneity of multi-element mixing, and enhanced catalytic performance, which are promising for use in energy and electronic fields. Meanwhile, the one-step, continuous, and scalable nature of the flame aerosol process makes it ideal for industrial production, paving the way to large-scale development and application of HEA nanomaterials.

Methods

Chemicals

RuCl₃•H₂O (35–40% Ru, Thermo Scientific), RhCl₃•H₂O (38 wt.% Rh, Thermo Scientific), PdCl₂ (59.5% Pd, Thermo Scientific), IrCl₃•3H₂O (53–56 % Ir, Thermo Scientific), PtCl₄ (99%, Acros Organics), OsCl₃•3H₂O (52–56% Os, Thermo Scientific), AuCl₃ (99%, Thermo Scientific), FeCl₂•4H₂O (99%, Acros Organics), CoCl₂•6H₂O (98%, Thermo Scientific), NiCl₂ (98%, Thermo Scientific), CuCl₂•2H₂O (98%, Thermo Scientific), ZnCl₂ (98%, Thermo Scientific), Ammonium tungsten oxide (NH₄)₆H₂W₁₂O₄₀•H₂O (99.9%, Thermo Scientific), Ga(NO₃)₃•H₂O (99.9%, Thermo Scientific), Ce(NO₃)₃•6H₂O (99.5%, Acros Organics), La(NO₃)₃•6H₂O (98 + %, Acros Organics), Mg(NO₃)₂•6H₂O (99 + %, Acros Organics), ZrCl₄ (99.5%, Alfa Aesar), sodium dodecyl sulfate (SDS, 99%, Acros Organics), tetraethyl orthosilicate (TEOS, 98%, Thermo Scientific), hexadecyltrimethylammonium bromide (CTAB, 99 + %, Acros Organics), HCl (37%, Acros Organics), and ethanol (200 proof, Decon Labs).

Support materials

Carbon substrate was purchased from fuel cell store (Ketjenblack 300, #NC 9849314); TiO₂ nanocrystal substrate was purchased from Sigma-Aldrich (anatase, nanopowder <25 nm, #637254-50 G); Carbon nanotube substrate was purchased from Cheap Tubes (8–15 nm outer diameter, #03040302); Graphene substrate was provided by HydroGraph (fractal dimension $D_f = 2.5 \pm 0.1$, #RGA-COOH-1);

UiO-66 MOF substrate was synthesized via the following procedures based on a previous study⁸⁸: 0.6 mmol of 1,4-benzenedicarboxylic acid (BDC, 99%, Thermo Scientific) and 1.2 mmol of triethylamine (TEA, Thermo Scientific) were dissolved in 140 mL of N,N-dimethylformamide (DMF, 99.8%, Fisher Chemical), and 20.62 mL of acetic acid (99.7%, Thermo Scientific) was added. The mixture was stirred at 24 °C room temperature for 10 minutes. Then, 0.6 mmol of ZrCl₄ was dissolved in 10 mL of DMF and added to the previous solution. The reaction precursor was kept at 120 °C for 6 h to grow the MOF

crystals. After that, the product was centrifuged, washed twice with DMF and once with methanol. Finally, the MOF was obtained by drying under vacuum at 60 °C overnight;

Fe-N-C substrate was synthesized via the following procedures according to our previous publication⁴⁷: Typically, the Fe₂O₃ doped ZIF-8 was prepared in methanol. Solution A was prepared by dispersing 10 mg Fe₂O₃ (~5 nm, Alfa Aesar) and 6.78 g Zn(NO₃)₂ in 150 mL methanol. Solution B was prepared by dissolving 7.88 g 2-methylimidazole in 150 mL methanol. Then, solution B was poured into solution A at 24 °C room temperature, followed by stirring for 20 minutes. Then, the mixture was kept at 60 °C for 24 h. The obtained Fe₂O₃ doped ZIF-8 was collected by centrifugation and washed with ethanol three times. Then, the obtained Fe₂O₃ doped ZIF-8 powder was dried in a vacuum oven at 60 °C for 12 h. The Fe-N-C catalyst was obtained by pyrolyzing 300 mg Fe₂O₃ doped ZIF-8 powder in forming gas (10% H₂/Ar) at 1100 °C for 1 h.

Synthesis of supported HEA nanoparticles

In a typical synthesis of RuRhPdIrPt/carbon, aqueous solutions of RuCl₃, RhCl₃, PdCl₂, IrCl₃, and PtCl₄ were first prepared. The metal chloride solutions, carbon support and 200 mg SDS were added in distilled H₂O to achieve a total metal cation concentration of 5 mM, metal:carbon ratio of 40 wt.%, and total volume of 250 mL. Then, the mixture was ultrasonically treated with a probe sonicator for 10 min to form a homogeneous precursor and support suspension. In the flame reactor, an inverted diffusion flame (O₂ introduced at the center, with H₂ and N₂ in the surrounding annular region) was generated by igniting a gas mixture of 15 L/min H₂, 6 L/min N₂, and 3.5 L/min O₂. The gases after combustion (N₂, water vapor, and excess H₂) passed through a converging-diverging nozzle (drill Ø0.1111 inch, #34), forming a sonic velocity hot stream. When the temperature and pressure became stable, the suspension precursor was injected into the throat section of the nozzle through two ports (drill Ø0.1040 inch, #37) by a peristaltic pump at 250 mL/h flow rate. These operating conditions produce a typical temperature of ~720 °C and an in-situ reducing atmosphere in the reactor chamber. The precursor was instantly atomized into micro-droplets. At high temperature, droplets evaporated instantly, and the HEA nanoparticles formed on the carbon support. Downstream of the reactor chamber, the produced HEA/carbon was immediately quenched by a 100 L/min N₂ flow to stabilize the HEA single phase and prevent particle agglomeration and sintering. The product was collected on a filter membrane. A downstream vacuum pump provided the pressure drop across the filter. Finally, the product was washed 3 times with ethanol and dried in air at 100 °C overnight. The HEA/TiO₂ nanocrystal, HEA/carbon nanotube, HEA/graphene, HEA/UiO-66, and HEA/Fe-N-C were fabricated by the same procedures.

For the simultaneous formation of HEA nanoparticles and mesoporous silica support, in a typical synthesis of RuRhPdIrPt/mesoporous silica, the RuCl₃, RhCl₃, PdCl₂, IrCl₃, and PtCl₄ aqueous solutions for HEA and the TEOS, CTAB, and HCl for mesoporous silica support were dissolved in a mixed solvent of 60 mL ethanol and 240 mL H₂O, to form a solution precursor but not a suspension precursor, with a TEOS concentration of 90 mM, CTAB:TEOS molar ratio of 0.25, and pH of ~2. The designed metal:SiO₂ ratio varied from 1 wt.% to 50 wt.%. The flame aerosol synthesis procedures were the same as those described above. The intermediate product collected on a filter membrane was washed 3 times with ethanol to remove the micelle template and generate mesopores, then dried in air at 100 °C overnight.

Material characterizations

High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) imaging with elemental mapping by energy dispersive x-ray spectroscopy (EDS), and live fast-Fourier-transform (FFT) patterns were obtained using a JEOL 2100-F 200 kV field-emission analytical transmission electron microscope; Aberration

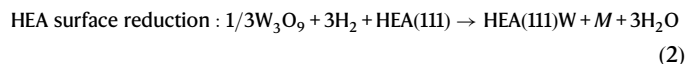
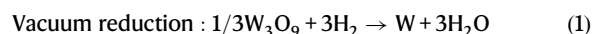
corrected scanning transmission electron microscopy (AC-STEM) was carried out on a JEOL JEM-ARM200F thermal-field emission microscope with a probe spherical aberration (Cs) corrector working at 200 kV. A JEOL JEM 2010 Transmission Electron Microscope (TEM) was also used to observe the materials. The particle size distribution curve was obtained by statistically analyzing 100 HEA nanoparticles dispersed on different substrate grains. A Cross-Beam Focused Ion Beam-Scanning Electron Microscopy (FIB-SEM) Workstation (Carl Zeiss AURIGA) also was used to observe the morphology of substrates; X-ray diffraction (XRD) patterns were acquired using an X-ray diffractometer (Rigaku Ultima IV) with Cu K α source ($\lambda = 0.15418$ nm). The diffractometer was operated at 40 mA and 40 kV at a scanning rate of 2°/min; The ex-situ X-ray absorption spectroscopy (XAS) spectra were collected at 8-ID beamline (Inner Shell Spectroscopy) of Brookhaven Lab's National Synchrotron Light Source II. X-ray photoelectron spectroscopy (XPS, Thermo Fisher K-Alpha Plus) measurements were conducted to analyze the surface metal state, with surface etching time of 0 s, 120 s, and 240 s, respectively. The photoelectron spectrometer system was configured with an Al K α excitation source with spot size of 400 μ m; The inductively coupled plasma optical emission spectrometer (ICP-OES, Thermo Scientific iCAP 6000) and X-ray fluorescence (XRF, Epsilon 1, Malvern Panalytical) were used to measure the HEA elemental composition; N₂ physisorption measurements (Micromeritics Tri-Star II) were used to characterize surface area and pore structure of the materials. Reported specific surface area was calculated based on the Brunauer-Emmett-Teller (BET) method, and the pore size distribution were calculated by the Barrett-Joyner-Halenda (BJH) or Horvath-Kawazoe methods based on the isotherms; Fourier transform infrared (FTIR) spectra was collected using a Bruker Vertex 70 Spectrometer; Raman spectra were measured by a Renishaw In Via spectrometer with 514 nm excitation laser focused through a 20 $\times$ microscope objective; H₂ temperature programmed reduction (H₂-TPR) was conducted using an Auto Chem II 2920 apparatus (Micromeritics) with 10% H₂ in Ar and a flow rate of 50 mL/min. The material was heated up to 850 °C with a 10 °C/min ramp rate.

DFT calculations

Calculations of surface energy and O desorption were performed using the Vienna Ab initio Simulation Package (VASP)^{89–92}. The PBE exchange correlation functional was used⁹³, with a plane wave energy cutoff of 520 eV. The O, Ir, Pd, Pt, Au, Os, Rh, Ru, Zr, W, and Ce projector augmented wave (PAW) pseudopotentials were chosen^{94–96}. All calculations were spin-polarized, with magnetic moments initialized ferromagnetically. Relaxations were performed with a force convergence criterion of 0.02 eV/Å and final static calculations were run with the linear tetrahedron method enabled to obtain accurate energies⁹⁷. Γ -centered Monkhorst-Pack k-point meshes were used to sample the Brillouin zone⁹⁸. Bulk metal and HEA structures were fully relaxed to obtain their equilibrium lattice parameters, which were then used to construct surface slabs with vacuum gaps of at least 10 Å. For the HEA case, a 125-atom special quasirandom structure (SQS) of bulk RuRhPdIrPt and PdPtIrOsAu consisting of five FCC (111) layers was generated using the Alloy Theoretic Automated Toolkit (ATAT)^{99–102}. For surface energy calculations, three different (111) surfaces were calculated for each HEA. For O adsorption calculations, this was then cleaved to construct a five-layer slab exposing a surface layer with a near-equiatom composition of Ir₄Pd₅Pt₅Rh₅Ru₆. For the bulk metal relaxations, k-meshes were automatically generated with a density of 30 Å along each reciprocal lattice vector, while for the HEA structure, a 2 $\times$ 2 $\times$ 2 mesh was used. Details of the surface slab structures are provided in Additional Supplementary Files. After applying a full coverage of O atoms to one surface (and, in the HEA case, substituting one of the surface metal sites), the ions were relaxed before and after removing a single O atom to obtain the O desorption energy, given by $E_{\text{des}} = E_{\text{slab-O}} + E_{\text{O}} - E_{\text{slab}}$, where E_{slab} and $E_{\text{slab-O}}$ are the energies

before and after removing the O atom, respectively, and E_{O} is the energy of a single O atom (calculated in a box of side length 8 Å with a 3 $\times$ 3 $\times$ 3 k-point mesh). In the HEA case, various sites were sampled from the surface for Zr/W/Ce substitution, along with sampling adjacent O sites for desorption. We sampled 64, 52, and 15 desorption configurations for Zr, W, and Ce, respectively. Structures were visualized using VESTA¹⁰³.

Also, we investigated the Gibbs energies of reduction of W/Zr/Ce oxide clusters in two cases. Taking W as an example⁷⁰:



where HEA(111)W and HEA(111) are HEA surfaces with and without a single W atom substituted (M in this case is the surface atom replaced by W). W and M are taken to be in their corresponding bulk metals, while the molecular species (H₂, and H₂O) are gaseous. For Zr and Ce, we have analogous reactions for the reduction of Zr₃O₆ and Ce₃O₆ but with 2 H₂/H₂O.

To compute reaction energies, the energies of each reactant/product were calculated from DFT. We used the same HEA surface structures used in our desorption calculations (excluding the oxygen layer), with a vacuum gap of at least 15 Å. For each element, all 25 HEA surface sites were sampled for substitution to obtain an average defect/reaction energy. Metal oxide cluster structures were obtained from literature^{104–106}. Cubic boxes of side length 25 Å and 15 Å were used for relaxations of the metal oxide clusters and H₂/H₂O, respectively. All settings from our desorption calculations were maintained, except that a DFT-D3 van der Waals correction¹⁰⁷ was used in combination with the PBE functional. For the molecules, vibrational modes were calculated using finite (central) differences without symmetry, as implemented in VASP.

To account for finite temperature vibrational effects, we calculated Gibbs free energies at $T = 800$ °C = 1073.15 K. For the molecules we used

$$G = E_{\text{DFT}} + E_{\text{ZP}} + H_{\text{vib}, T} - TS_{\text{vib}, T} \quad (3)$$

where E_{ZP} , $H_{\text{vib}, T}$, $S_{\text{vib}, T}$ are the zero-point energy, vibrational enthalpy, and vibrational entropy at temperature T obtained from the calculated vibrational modes using post-processing tools in VASPKIT¹⁰⁸. For the bulk metals we used

$$G = E_{\text{DFT}} + (H_T^\circ - H_0^\circ) - TS_T^\circ \quad (4)$$

where $H_T^\circ - H_0^\circ$ and S_T° are the enthalpy (relative to $T = 0$) and entropy obtained from available thermodynamic data (linearly interpolating between data at 1000 K and 1100 K when necessary)^{109–116}. No finite temperature term was added to the HEA surface energies, since these appear on opposite sides of the reduction reaction and should therefore cancel if we assume the entropy difference from substituting a single surface atom to be negligible.

In addition, we investigated the HOR catalytic reaction. The adsorption energies of *H, *OH, and *H₂O on the (111) surface of the high-entropy alloy (HEA) and a reference PtRu alloy were calculated using four-layer surface slabs constructed from representative 64-atom SQSs. The lateral lattice constants were fixed to the equilibrium bulk values of each SQS. A vacuum space of 15 Å was introduced to avoid interactions between periodic images. The adsorbates were initially placed at their most stable sites, namely FCC hollow for *H, and atop for *OH and *H₂O, as determined from preliminary site tests. Adsorption was evaluated independently on both the top and bottom

slab surfaces to sample diverse local environments. During structural optimization, the bottom slab layer was fixed, while the top three layers and all adsorbates were fully relaxed. Free energy diagrams for HOR were then constructed by averaging the adsorption energies over all non-equivalent configurations across the two surfaces.

Electrochemical measurements

Electrochemical measurements were performed using a CHI 760e potentialstat. A PTFE container (100 mL) was used as the electrochemical cell to avoid the possible corrosion of a glass cell. A glassy carbon rotating disk electrode (5 mm in diameter), a saturated calomel electrode (SCE, saturated KCl), and a graphite rod were employed as the working electrode, the reference electrode, and the counter electrode, respectively. All potentials were converted to values with reference to a reversible hydrogen electrode (vs. RHE). The calibration was performed in H₂-saturated 0.1 M KOH using PtRu/C as a working electrode, and the potential at 0 mA cm⁻² was used to convert SCE to RHE potential.

The 0.1 M KOH solution was prepared by dissolving 5.6 g KOH in 1.0 L DI water, and the prepared electrolyte was stored in a plastic container. The pH is about 12.9 ± 0.1.

Catalyst ink was prepared by dispersing the electrocatalysts in a mixture of isopropanol and Nafion (5%) (v/v = 1:0.01, 2 mg_{catal} mL⁻¹) by sonication. 10 µL of ink was cast on the working electrode and dried under ambient conditions. The final PGM loadings were determined by ICP-OES (~10 µg_{PGM} cm⁻²).

$$\text{PGM loading}(\text{mg cm}^{-2}) = \frac{2\text{mg mL}^{-1} \cdot w_{\text{PGM}} \cdot 0.01\text{ mL}}{0.196\text{ cm}^{-2}} \quad (5)$$

Where w_{PGM} is the PGM loading in the catalyst obtained by ICP-OES. The ECSA value is calculated as shown below:

$$\text{ECSA}(\text{m}^2 \text{g}^{-1}) = \frac{Q_{\text{HUPD}}(\text{C})}{0.21\text{ mC cm}^{-2} \cdot \text{PGM loading}(\text{mg})} \cdot 10^5 \quad (6)$$

Where Q_{HUPD} is the hydrogen desorption charge obtained by integrating the area in CVs.

All of the measurements were carried out at 24 °C room temperature. The catalysts were activated by performing cyclic voltammetry (CV) in N₂-saturated 0.1 M KOH solutions (50 mL) at a scan rate of 200 mV/s between 0–1.0 V (vs. RHE). The HOR measurements were carried out with a rotating disk electrode (RDE) in H₂-saturated 0.1 M KOH at a rotation speed of 1600 rpm and a sweep rate of 10 mV s⁻¹. Kinetic currents were calculated using the Koutecky-Levich equation:

$$i_k = \frac{i_l \cdot i_{0.05}}{i_l - i_{0.05}} \quad (7)$$

Where i_k is the kinetic current, i_l limited current and $i_{0.05}$ current at 0.05 V after iR-correction.

Exchange current density (j_0) was calculated from the Butler-Volmer equation:

$$j_k = j_0 \left[e^{\alpha \frac{F}{RT}} - e^{\eta \frac{(1-\alpha)F}{RT}} \right] \quad (8)$$

Where α , R , T , and η represent the transfer coefficient, the universal gas constant (8.314 J mol⁻¹ K⁻¹), the operating temperature (297 K in this work), and the overpotential, respectively.

Accelerated durability test was performed at 24 °C room temperature by potential cycles between -0.1 and 0.4 V in H₂-saturated 0.1 M KOH at a scan rate of 100 mV/s.

The resistance of the electrochemical cell was evaluated by the iR compensation (98%) on a CHI 760e potentialstat, where the resistance values in 0.1 M KOH are ~46.3 ± 2.4 Ω.

Data availability

The data that support the findings and conclusions generated in this study are provided in the main article and the Supplementary Information. The calculation information is provided in the Supplementary Data 1 file. Source data are provided with this paper.

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Author contributions

S.L., M.T.S. and C.D. conceived the idea and experiments; S.L. designed all materials; S.L., K.T., K.C., and M.A.K. performed the materials synthesis; S.L., Q.J., D.W., Z.X., C.S., and C.D. performed the materials characterizations; J.L. performed the electrochemical tests and analysis; J.L.K., H.S., S.K., and W.C. performed DFT calculations and theoretical analysis; S.L. wrote the first draft of the manuscript; J.L., S.K., W.C., G.W., J.J.U., M.T.S., and C.D. revised the manuscript; S.L. and J.L. contributed equally; G.W., J.J.U., M.T.S., and C.D. jointly supervised this work.

Competing interests

The authors declare no competing interests.

Additional information

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